

Solutions to Linear Algebra Problems

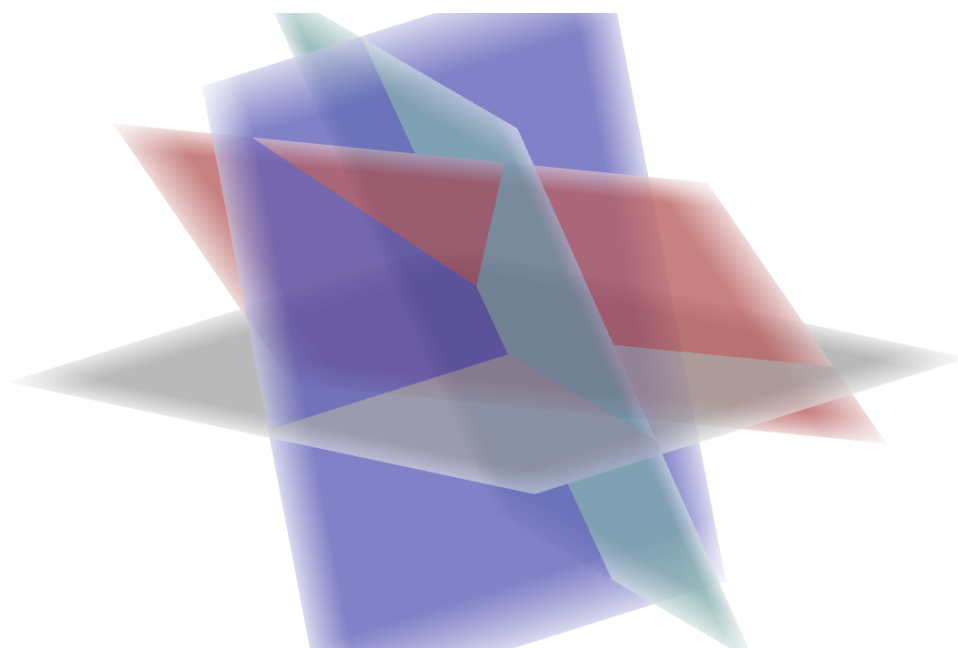
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Abstract

This note collects my solutions to selected problems in Linear Algebra. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- Linear Algebra Done Right 4e, Sheldon Axler



Problem 1.

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Source: *Linear Algebra Done Right* - Ex.12 p.25

Solution:

Let V_1, V_2 be two subspaces of V .

($\Rightarrow$): Suppose that $V_1 \cup V_2$ is a subspace of V . Suppose for contradiction that one of the subspaces V_1 or V_2 is not contained in the other. We can choose u, w such that $u \in V_1, u \notin V_2$ and $w \in V_2, w \notin V_1$.

Since $V_1 \cup V_2$ is a subspace, we have $u + w \in V_1 \cup V_2$. This means $u + w$ must be in V_1 or V_2 .

If $u + w \in V_1$, since V_1 is a subspace and $-u \in V_1$, then $w = (u + w) - u \in V_1$, which is a contradiction. Similarly, if $u + w \in V_2$, since V_2 is a subspace and $-w \in V_2$, then $u = (u + w) - w \in V_2$, which is also a contradiction.

Therefore, $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.

($\Leftarrow$): If $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$, then $V_1 \cup V_2 = V_2$ or $V_1 \cup V_2 = V_1$, which is a subspace.

Problem 2.

Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that the list $v_1, \dots, v_m$ is linearly independent if and only if the list $w_1, \dots, w_m$ is linearly independent.

Source: *Linear Algebra Done Right* - Ex.14 p.38

Solution:

($\Rightarrow$): Suppose the list $v_1, \dots, v_m$ is linearly independent. Consider a combination of $w_1, \dots, w_m$ such that:

$$\begin{aligned} c_1 w_1 + c_2 w_2 + \dots + c_m w_m &= 0 \\ \Rightarrow c_1 v_1 + c_2(v_1 + v_2) + \dots + c_m(v_1 + v_2 + \dots + v_m) &= 0 \\ \Rightarrow (c_1 + c_2 + \dots + c_m)v_1 + (c_2 + \dots + c_m)v_2 + \dots + c_m v_m &= 0 \end{aligned}$$

Since $v_1, \dots, v_m$ are linearly independent, it follows that $c_1 = c_2 = \dots = c_m = 0$. Therefore, the list $w_1, \dots, w_m$ are linearly independent.

($\Leftarrow$): Suppose $w_1, \dots, w_m$ are linearly independent. Consider a combination of $v_1, \dots, v_m$ such

that:

$$\begin{aligned}c_1v_1 + c_2v_2 + \cdots + c_mv_m &= 0 \\ \Rightarrow c_1w_1 + c_2(w_2 - w_1) + c_3(w_3 - w_2) + \cdots + c_m(w_m - w_{m-1}) &= 0 \\ \Rightarrow (c_1 - c_2)w_1 + (c_2 - c_3)w_2 + \cdots + (c_{m-1} - c_m)w_{m-1} + c_mw_m &= 0\end{aligned}$$

Since $w_1, \dots, w_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $v_1, \dots, v_m$ are linearly independent.

Problem 3.

Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Source: Linear Algebra Done Right - Ex.5 p.43

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of U and $w_1, w_2, \dots, w_n$ be a basis of W .

Since $V = U + W$, we have the list $\{u_1, u_2, \dots, u_m, w_1, w_2, \dots, w_n\}$ spans V .

Applying the procedure in Theorem 2.30 to reduce this list to a basis of V produces a basis consisting of the vectors $u_1, \dots, u_m$ and some of the w 's (none of the u 's get deleted in this procedure because $u_1, \dots, u_m$ is linearly independent).

Problem 4.

Suppose V is finite-dimensional and U is a subspace of V with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of V , each of dimension $n - 1$, whose intersection equals U .

Source: Linear Algebra Done Right - Ex.16 p.49

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of U . This list can be extended to a basis of V by adjoining $n - m$ additional vectors (by Theorem 2.32), say $w_1, w_2, \dots, w_{n-m}$. We denote this extended list as L .

Let V_i ($1 \leq i \leq n - m$) be the subspace spanned by L without including w_i . Clearly $\dim V_i = n - 1$ and $\bigcap_{i=1}^{n-m} V_i$ is also a subspace that contains U .

Consider $v \in \bigcap_{i=1}^{n-m} V_i$. We have:

$$\begin{aligned} a_1u_1 + a_2u_2 + \cdots + a_mu_m + b_2w_2 + b_3w_3 + \cdots \\ + b_{n-m}w_{n-m} = v \quad (\text{since } v \in V_1) \end{aligned}$$

$$\begin{aligned} c_1u_1 + c_2u_2 + \cdots + c_mu_m + d_1w_1 + d_3w_3 + d_4w_4 \\ + \cdots + d_{n-m}w_{n-m} = v \quad (\text{since } v \in V_2) \end{aligned}$$

By subtracting, we have:

$$\begin{aligned} (a_1 - c_1)u_1 + (a_2 - c_2)u_2 + \cdots + (a_m - c_m)u_m - d_1w_1 \\ + (b_2 - d_2)w_2 + b_3w_3 + \cdots + (b_{n-m} - d_{n-m})w_{n-m} = 0 \end{aligned}$$

Since L is a list of independent vectors and by repeating this process, this implies that all coefficients for w_i 's are 0 and v is uniquely represented in the basis of U .

We conclude that $\bigcap_{i=1}^{n-m} V_i = U$.

Problem 5.

Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V . In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

Source: *Linear Algebra Done Right* - Ex.13 p.58

Solution:

Let U be a subspace of V , let $S \in \mathcal{L}(U, W)$, and let $u_1, u_2, \dots, u_n$ be a basis of U . By Theorem 2.32, this list can be extended to a basis of V :

$$u_1, u_2, \dots, u_n, v_{n+1}, v_{n+2}, \dots, v_m.$$

For $1 \leq i \leq n$, let

$$w_i = Su_i.$$

Choose arbitrary vectors $w_{n+1}, w_{n+2}, \dots, w_m \in W$. By the linear map lemma, there exists a unique $T \in \mathcal{L}(V, W)$ such that

$$Tu_1 = w_1 = Su_1, \quad \dots, \quad Tu_n = w_n = Su_n,$$

and

$$Tv_{n+1} = w_{n+1}, \quad \dots, \quad Tv_m = w_m.$$

Now let $u \in U$. Since $u_1, \dots, u_n$ is a basis of U , there exist scalars $a_1, \dots, a_n$ such that

$$u = a_1u_1 + a_2u_2 + \dots + a_nu_n.$$

Thus

$$\begin{aligned} Tu &= T(a_1u_1 + a_2u_2 + \dots + a_nu_n) \\ &= a_1Tu_1 + a_2Tu_2 + \dots + a_nTu_n \\ &= a_1Su_1 + a_2Su_2 + \dots + a_nSu_n \\ &= S(a_1u_1 + a_2u_2 + \dots + a_nu_n) \\ &= Su. \end{aligned}$$

Therefore, $Tu = Su$ for all $u \in U$, so T is an extension of S to V .

Problem 6.

Suppose V and W are finite-dimensional and that U is a subspace of V . Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\text{null } T = U$ if and only if $\dim U \geq \dim V - \dim W$.

Source: Linear Algebra Done Right - Ex.18 p.67

Solution:

($\Rightarrow$) Suppose there exists $T \in \mathcal{L}(V, W)$ such that $\text{null } T = U$. By the fundamental theorem of linear maps, we have:

$$\dim V = \dim \text{null } T + \dim \text{range } T$$

$$\Rightarrow \dim U = \dim V - \dim \text{range } T \geq \dim V - \dim W$$

($\Leftarrow$) Suppose U is a subspace of V such that $\dim U \geq \dim V - \dim W$. Let $u_1, u_2, \dots, u_n$ be a basis of U . This list can be extended to $u_1, u_2, \dots, u_n, v_1, v_2, \dots, v_m$ to be a basis of $V \Rightarrow \dim V = n + m$. We have $\dim U \geq \dim V - \dim W \Rightarrow \dim W \geq \dim V - \dim U = m$. Therefore, we can let $w_1, w_2, \dots, w_m$ be an independent list of vectors in W . Define $T : V \rightarrow W$ as:

$$T(c_1u_1 + \dots + c_nu_n + d_1v_1 + \dots + d_mv_m) = d_1w_1 + d_2w_2 + \dots + d_mv_m$$

It is easy to verify that T is a linear map by checking $T(u + v) = T(u) + T(v)$ and $T(\lambda u) = \lambda T(u)$, $\forall u, v \in V$.

We have for all $u \in U$, $Tu = T(c_1u_1 + \dots + c_nu_n) = 0w_1 + \dots + 0w_m = 0 \Rightarrow u \in \text{null } T \Rightarrow U \subseteq \text{null } T$.

Now, let $u \in \text{null } T$. We have $Tu = 0 \Rightarrow d_1 = d_2 = \dots = d_m = 0$ since $w_1, w_2, \dots, w_m$ are

independent. Thus, u can be expressed as $c_1u_1 + c_2u_2 + \cdots + c_nu_n \Rightarrow u \in U \Rightarrow \text{null } T \subseteq U$.
Therefore, $U = \text{null } T$.

Problem 7.

Suppose there exists a linear map on V whose null space and range are both finite-dimensional. Prove that V is finite-dimensional.

Source: *Linear Algebra Done Right* - Ex.15 p.66

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of $\text{null } T$ and $v_1, v_2, \dots, v_n$ be a basis of $\text{range } T$. Let $w_1, w_2, \dots, w_n \in V$ such that $v_1 = Tw_1, v_2 = Tw_2, \dots, v_n = Tw_n$.

For any $z \in V$, we have:

$$\begin{aligned} Tz &= a_1Tw_1 + a_2Tw_2 + \cdots + a_nTw_n \\ &= T(a_1w_1 + a_2w_2 + \cdots + a_nw_n) \end{aligned}$$

Let $u = z - a_1w_1 - a_2w_2 - \cdots - a_nw_n \Rightarrow Tu = 0 \Rightarrow u \in \text{null } T$

$$\Rightarrow u = b_1u_1 + b_2u_2 + \cdots + b_mu_m$$

$$\Rightarrow z = a_1w_1 + a_2w_2 + \cdots + a_nw_n + b_1u_1 + b_2u_2 + \cdots + b_mu_m$$

This implies that the list $u_1, u_2, \dots, u_m, w_1, w_2, \dots, w_n$ spans V .

Therefore, V is finite dimensional.

Problem 8.

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that there exist a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ are 0 except that the entries in row k , column k , equal 1 if $1 \leq k \leq \dim \text{range } T$.

Source: *Linear Algebra Done Right* - Ex.5 p.79

Solution:

By Thm 3.21: $\dim V = \dim \text{null } T + \dim \text{range } T$

Let $m = \dim \text{null } T = \dim V - \dim \text{range } T$

Let $u_1, u_2, \dots, u_m$ be a basis of $\text{null } T$. Since $\text{null } T$ is a subspace of V , this list can be extended to $v_1, v_2, \dots, v_n, u_1, \dots, u_m$ to be a basis of V . This implies $\dim V = m + n$ and $\dim \text{range } T = n$.

Consider $a_1Tv_1 + a_2Tv_2 + \cdots + a_nTv_n = 0$.

$$\Rightarrow T(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = 0$$

$$\Rightarrow a_1v_1 + \cdots + a_nv_n \in \text{null } T$$

$$\Rightarrow a_1v_1 + \cdots + a_nv_n = b_1u_1 + \cdots + b_mu_m$$

$$\Rightarrow a_1 = a_2 = \cdots = a_n = 0 \text{ since } v_1, v_2, \dots, v_n, u_1, \dots, u_m \text{ are independent}$$

This means $Tv_1, Tv_2, \dots, Tv_n$ are independent in W and this list can be extended to become a basis of W . With this construction of a basis of V and a basis of W , $\mathcal{M}(T)$ will have the desired entries.

Problem 9.

Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \text{range } ST \leq \min\{\dim \text{range } S, \dim \text{range } T\}.$$

Source: Linear Algebra Done Right - Ex.23 p.67

Solution:

We have $\text{range } ST$ is a subspace of $\text{range } S$. Therefore,

$$\dim \text{range } ST \leq \dim \text{range } S.$$

Note that the restriction of S to $\text{range } T$, denoted $S|_{\text{range } T}$, is in $\mathcal{L}(\text{range } T, W)$. By Thm 3.21, we have:

$$\begin{aligned} \dim \text{range } T &= \dim \text{null}(S|_{\text{range } T}) + \dim \text{range}(S|_{\text{range } T}) \\ &= \dim \text{null}(S|_{\text{range } T}) + \dim \text{range } ST \\ &\geq \dim \text{range } ST \end{aligned}$$

Therefore,

$$\dim \text{range } ST \leq \min\{\dim \text{range } S, \dim \text{range } T\}.$$